

Experiment-7

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Code:

```
#include <iostream>
#include <vector>
#include <list>
using namespace std;
int main()
{
    vector<int> v;
    v.push_back(10);
    v.push_back(20);
    v.push_back(30);
    cout << "Vector Elements: ";
    for(int i = 0; i < v.size(); i++)
        cout << v[i] << " ";
    cout << endl;

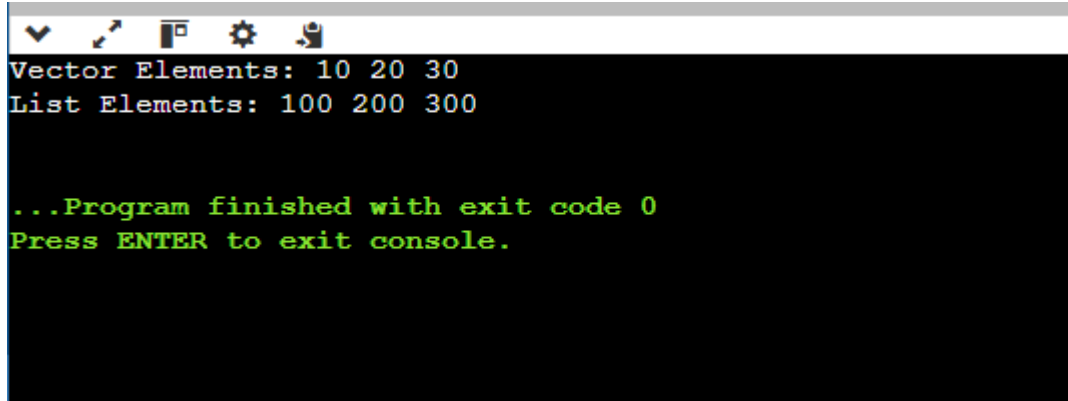
    list<int> l; // changed variable name from 1 to l
    l.push_back(100);
    l.push_back(200);
    l.push_back(300);
    cout << "List Elements: ";
    for(auto it = l.begin(); it != l.end(); it++)
```

```

    cout << *it << " ";
cout << endl;
return 0;}

```

output:



A screenshot of a console window with a black background and white and green text. The window has a title bar with standard OS icons. The output text is as follows:

```

Vector Elements: 10 20 30
List Elements: 100 200 300

...Program finished with exit code 0
Press ENTER to exit console.

```

scenario 1:

```

#include <iostream>
#include <vector>
#include <algorithm>

int main() {
    std::vector<int> marks = {45, 82, 33, 91, 56, 78, 49, 88};
    std::sort(marks.begin(), marks.end(), std::greater<int>());

    marks.erase(std::remove_if(marks.begin(), marks.end(), [](int m) {
        return m < 50;
    }), marks.end());

    std::cout << "--- Final Rank List (Passing Students) ---" << std::endl;
    if (marks.empty()) {
        std::cout << "No students passed the exam." << std::endl;
    } else {
        for (size_t i = 0; i < marks.size(); ++i) {
            std::cout << "Rank " << i + 1 << ": " << marks[i] << std::endl;
        }
    }
}

```

```

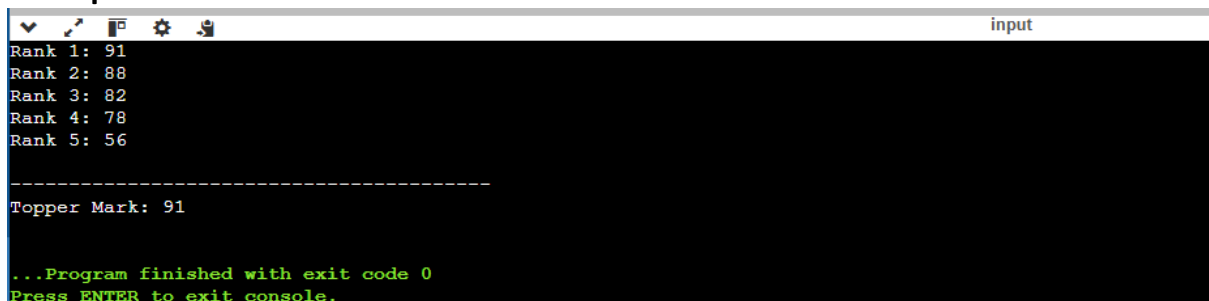
    }

    std::cout << "\n-----" << std::endl;
    if (!marks.empty()) {
        std::cout << "Topper Mark: " << marks[0] << std::endl;
    }

    return 0;
}

```

Output:



The screenshot shows a terminal window titled 'input'. The output of the program is as follows:

```

Rank 1: 91
Rank 2: 88
Rank 3: 82
Rank 4: 78
Rank 5: 56

-----
Topper Mark: 91

...Program finished with exit code 0
Press ENTER to exit console.

```

Scenario:2

```

#include <iostream>

#include <list>

#include <iterator>

using namespace std;

int main() {
    list<int> passengerIDs;

    passengerIDs.push_back(101);
    passengerIDs.push_back(102);
    passengerIDs.push_back(103);
    passengerIDs.push_back(104);
    passengerIDs.push_back(105);
}

```

```

passengerIDs.push_back(106);

auto it = next(passengerIDs.begin(), 2);
int vipID = 999;
passengerIDs.insert(it, vipID);

int cancelledID = 104;
passengerIDs.remove(cancelledID);

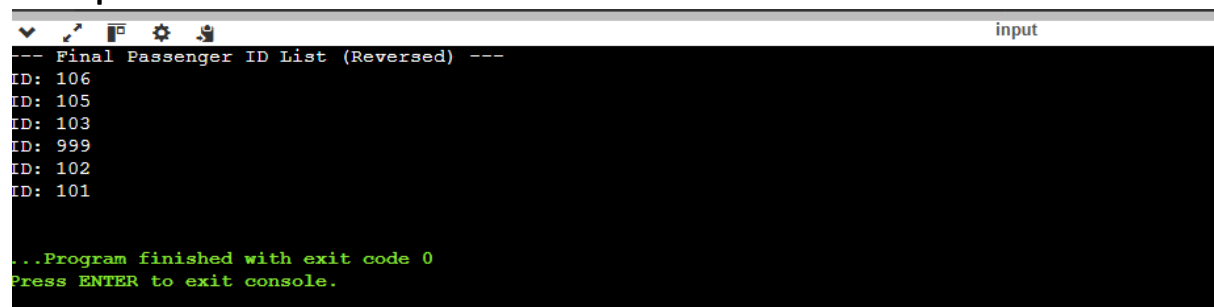
passengerIDs.reverse();

cout << "--- Final Passenger ID List (Reversed) ---" << endl;
for (int id : passengerIDs) {
    cout << "ID: " << id << endl;
}

return 0;
}

```

Output:



```

input
--- Final Passenger ID List (Reversed) ---
ID: 106
ID: 105
ID: 103
ID: 999
ID: 102
ID: 101

...Program finished with exit code 0
Press ENTER to exit console.

```

Scenario:3

```

#include <iostream>

#include <vector>

#include <algorithm> // For sort, unique, and remove_if
#include <numeric>   // For accumulate

```

```
using namespace std;
```

```
int main() {
```

```
    vector<double> cart;
```

```
    int numItems;
```

```
    cout << "Enter the number of products: ";
```

```
    cin >> numItems;
```

```
    cout << "Enter the prices: " << endl;
```

```
    for (int i = 0; i < numItems; i++) {
```

```
        double price;
```

```
        cin >> price;
```

```
        cart.push_back(price);
```

```
    }
```

```
    sort(cart.begin(), cart.end());
```

```
    auto last = unique(cart.begin(), cart.end());
```

```
    cart.erase(last, cart.end());
```

```
    for (double &price : cart) {
```

```
        price *= 0.90;
```

```
    }
```

```
    double total = accumulate(cart.begin(), cart.end(), 0.0);
```

```
    cout << "\nTotal Payable Amount after discount: " << total << endl;
```

```
    cart.erase(remove_if(cart.begin(), cart.end(), [](double p) {
```

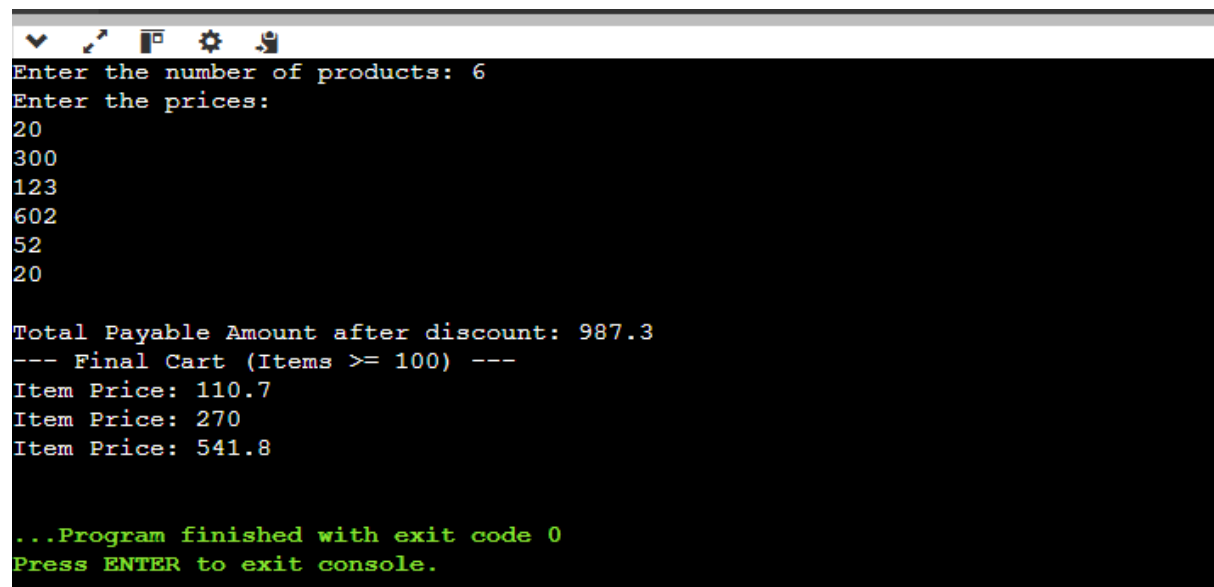
```
        return p < 100.0;
```

```
    }), cart.end());
```

```
    cout << "--- Final Cart (Items >= 100) ---" << endl;
    for (double price : cart) {
        cout << "Item Price: " << price << endl;
    }

    return 0;
}
```

Output:

A screenshot of a terminal window with a dark background and light-colored text. The window has a title bar with standard icons. The output shows the program's execution flow: input for the number of products (6), input for prices (20, 300, 123, 602, 52, 20), calculation of the total payable amount (987.3), and a list of item prices from the final cart (110.7, 270, 541.8). The program ends with a message indicating it finished with exit code 0 and a prompt to press ENTER to exit the console.

```
Enter the number of products: 6
Enter the prices:
20
300
123
602
52
20

Total Payable Amount after discount: 987.3
--- Final Cart (Items >= 100) ---
Item Price: 110.7
Item Price: 270
Item Price: 541.8

...Program finished with exit code 0
Press ENTER to exit console.
```